

Lab 1: Problem Solving in C/C++

1. Even or Odd

Input an integer and check whether it is even or odd.

2. Find the Maximum of Two Numbers

Take two numbers and print the larger one.

3. Sum of First N Natural Numbers

Input N and print the sum from 1 to N.

4. Check if a Number is Prime

Input a number and check whether it is prime or not.

5. Factorial of a Number

Input a number and calculate its factorial.

6. Reverse a Number

Input a number and print its reverse.

Example: 123 -> 321

7. Check if a Number is Palindrome

Check if a number reads the same forward and backward.

Example: 121 -> Palindrome

8. Print First N Fibonacci Numbers

Input N and print the first N Fibonacci numbers.

9. Count Digits of a Number

Input a number and count how many digits it has.

Example: 4567 -> 4 digits

10. Sum of Digits of a Number

Input a number and print the sum of its digits.

Example: 1234 -> $1+2+3+4 = 10$

11. Bubble sort

Input an unsorted array of size N and sort using bubble sort Example: 11 9

3 15 2 2 3 9 11 15

12. Linear Search

Input an N size array and find an element from the array using linear search. Example:
1 3 8 5 4, search key=4

13. Positive, Negative, or Zero

Input an integer and check whether it is positive, negative, or zero.

14. Swap Two Numbers

Input two numbers and swap their values.

Example: $a=5, b=8 \rightarrow a=8, b=5$

15. Simple Calculator

Input two numbers and an operator (+ - * /) and perform the operation.

16. Print Even Numbers from 1 to N

Input N and print all even numbers between 1 and N.

17. Find Second Largest Number

Input N numbers and print the second largest value.

18. Find Average of Array Elements

Input an array and calculate the average of all elements.

19. Remove Duplicates from Array

Input an array and print unique elements only.

20. Convert Lowercase to Uppercase Without Library Function

Input a string and manually convert lowercase letters to uppercase.